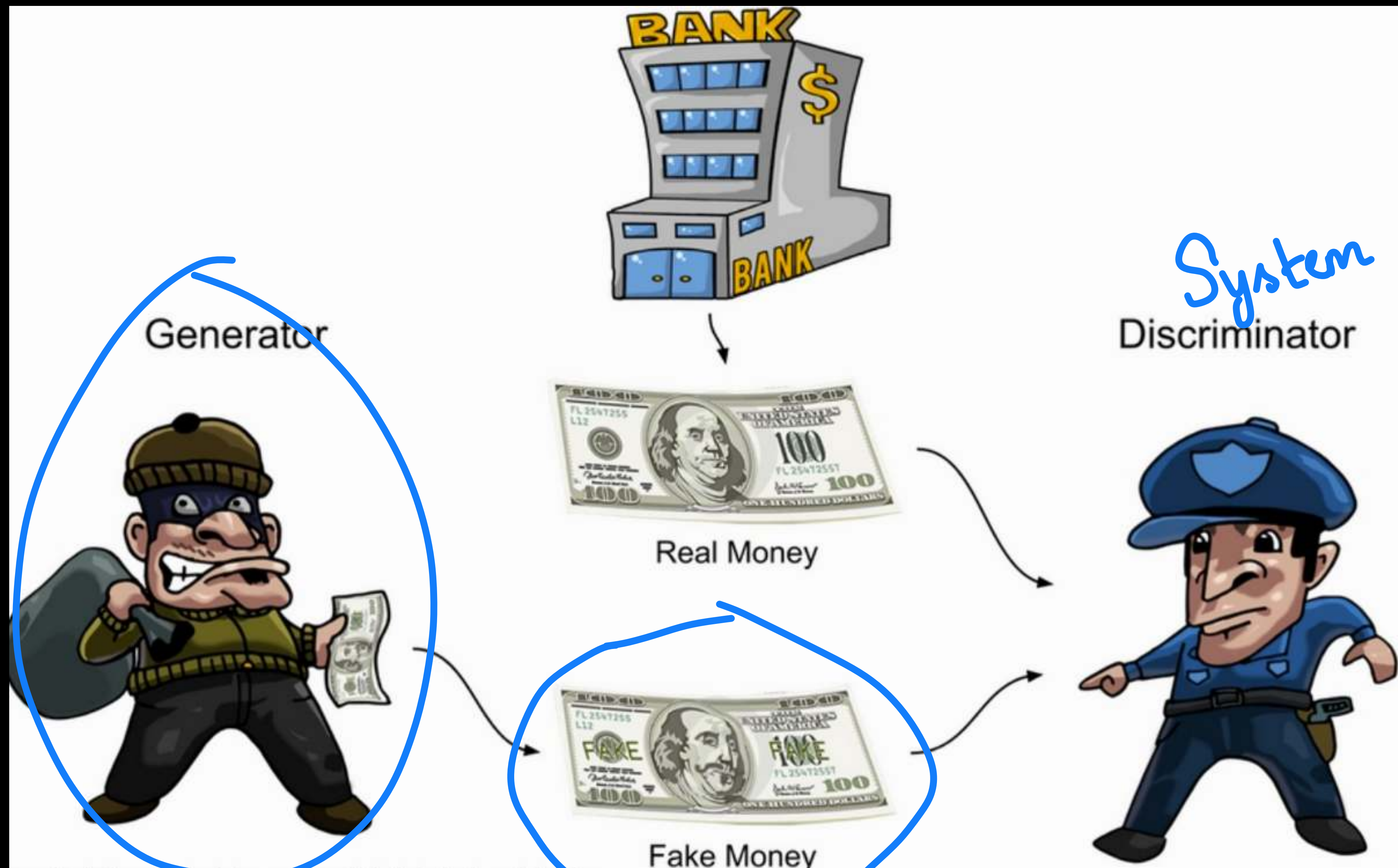


E9: 309 ADL 09-12-2020

GAN - Overview



Generative Adversarial Networks

* Generator networks

→ Differentiable network

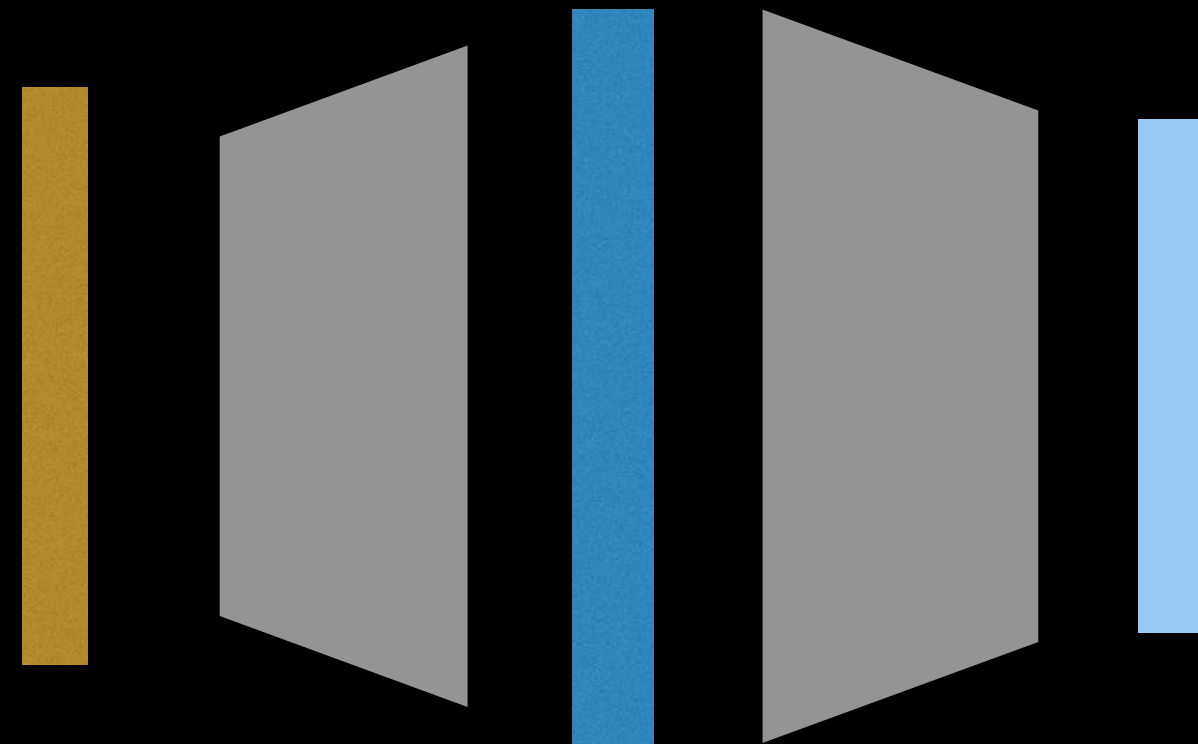
→ Draws a random noise sample

✓ Convert the noise to the data distribution

θ^G

$$\mathbf{x}' \sim p_{\theta^G}(\mathbf{x}|\mathbf{z})$$

$$\mathbf{z} \sim p(\mathbf{z})$$



$$\mathbf{x}' = G(\mathbf{z}, \theta^G)$$

Generative Adversarial Networks

* Discriminator network

→ Two class classifier

✓ Label 0 for x'

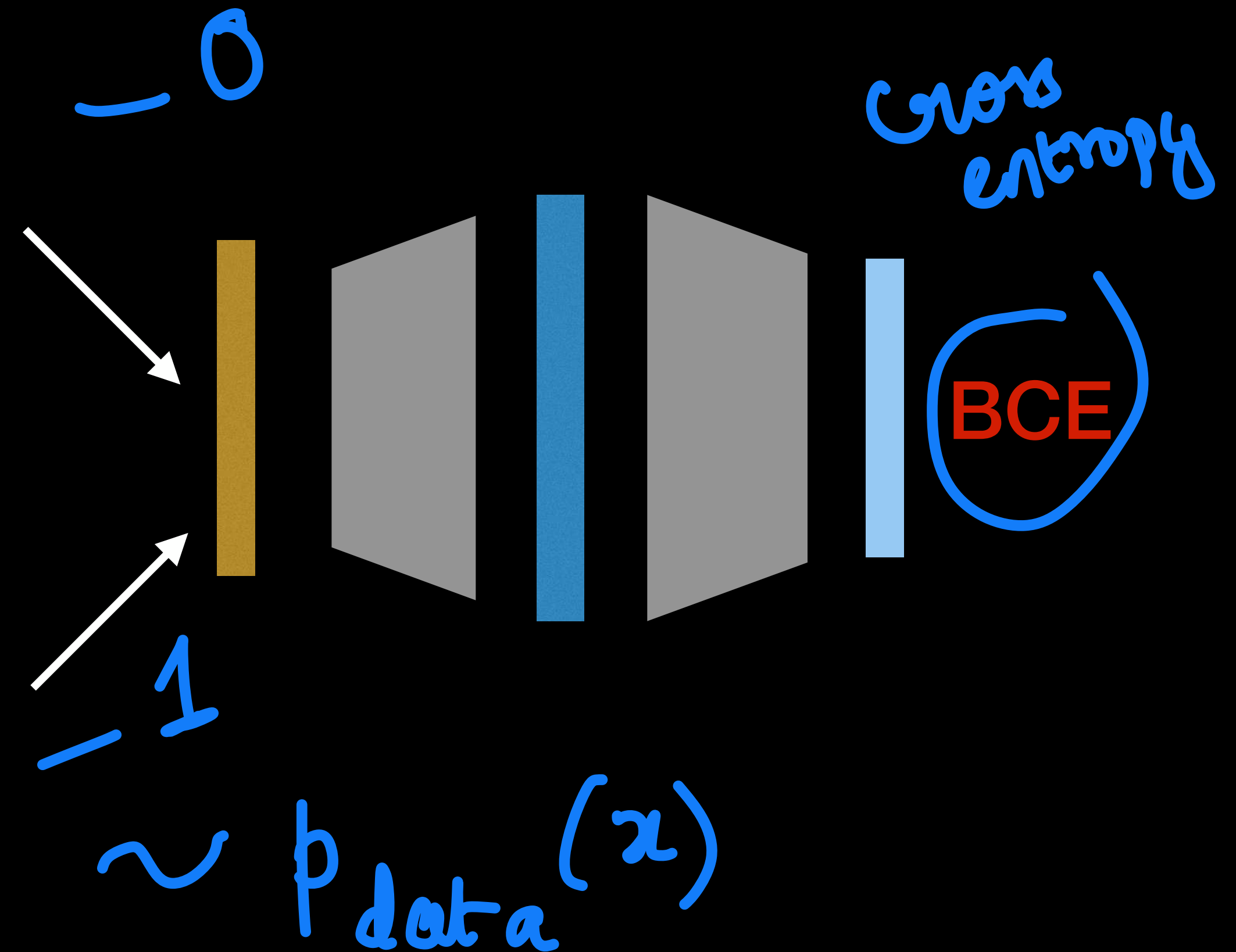
✓ Label 1 for x

→ Model learning

Model generated samples



True samples



Min-max game

✳ The binary cross entropy loss at the discriminator network is

$$E_D = -\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log D(G(\mathbf{z}))]$$

$\underbrace{\hspace{15em}}_{\theta^G, \theta^D}$

$E_G = -E_D$

class 1

$(1 - D(G(\mathbf{z})))$

class 0

✳ The discriminator is trained using B.C.E loss

$$\theta_D^n = \theta_D^{n-1} - \eta \frac{\partial E_D}{\partial \theta_D}$$



$$E = \sum_i 1 \underbrace{\log(D(x_i))}_{\text{class 1}} - \underbrace{(1 - 0) \log(1 - D(x_i))}_{\text{class 0}}$$

Min-max game

✳ The binary cross entropy loss at the discriminator network is

$$E_D = -\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log D(G(\mathbf{z}))]$$

$$E_G = -E_D$$

✳ The discriminator is trained using B.C.E loss

$$\theta_G^n = \theta_G^{n-1} - \eta \frac{\partial E_G}{\partial \theta_G}$$



GAN - min-max game

* Discriminator cost function

$$E_D = \min \left(\underbrace{-\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log(D(\mathbf{x}))]}_{\text{real}} - \underbrace{\mathbb{E}_{\mathbf{z} \sim p_{model}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))]}_{\text{fake}} \right)$$

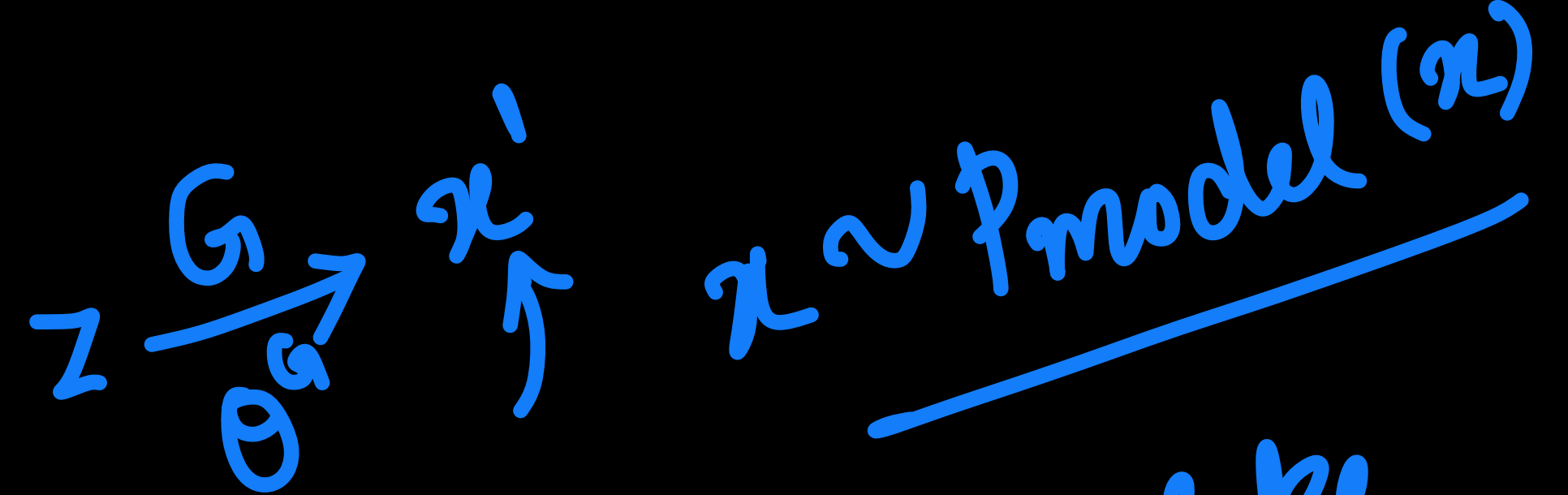
$$= \max \left(\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log(D(\mathbf{x}))] + \mathbb{E}_{\mathbf{x} \sim p_{model}(\mathbf{x})} [\log(1 - D(\mathbf{x}))] \right)$$

$p(\mathbf{z}) \leftarrow$ no parameters

* Optimal cost function

$$E^* = \min_G \max_D E(D, G)$$

$$E(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log(D(\mathbf{x}))] + \mathbb{E}_{\mathbf{x} \sim p_{model}(\mathbf{x})} [\log(1 - D(\mathbf{x}))]$$



real

fake

$p(z)$

$p(z) \leftarrow$ no parameters



GAN - min-max game

* Optimal discriminator

$$D^* = \operatorname{argmax}_D \left(\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log(D(\mathbf{x}))] + \mathbb{E}_{\mathbf{x} \sim p_{model}(\mathbf{x})} [\log(1 - D(\mathbf{x}))] \right)$$

$$= \operatorname{argmax}_D \int_{\mathcal{X}} \left(\log(D(\mathbf{x})) p_{data}(\mathbf{x}) + \log(1 - D(\mathbf{x})) p_{model}(\mathbf{x}) \right) d\mathbf{x}$$
$$= \operatorname{argmax}_D [\log(D(\mathbf{x})) p_{data}(\mathbf{x}) + \log(1 - D(\mathbf{x})) p_{model}(\mathbf{x})]$$

* Functional derivative

$$\frac{p_{data}(\mathbf{x})}{D^*(\mathbf{x})} - \frac{p_{model}(\mathbf{x})}{1 - D^*(\mathbf{x})} = 0 \quad \underline{\underline{D(\mathcal{X})}}$$

$\forall \mathbf{x}$

;

$$D^*(\mathbf{x}) = \frac{p_{data}(\mathbf{x})}{p_{data}(\mathbf{x}) + p_{model}(\mathbf{x})}$$

← optimal
← for a fixed generator



GAN - min-max game

★ Finding the best generator

optimal discriminator

$$G^* = \operatorname{argmin}_G \left(\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log(D^*(\mathbf{x}))] + \mathbb{E}_{\mathbf{x} \sim p_{model}(\mathbf{x})} [\log(1 - D^*(\mathbf{x}))] \right)$$

$$= \operatorname{argmin}_G \left(\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} \left[\log \frac{p_{data}(\mathbf{x})}{p_{data}(\mathbf{x}) + p_{model}(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_{model}(\mathbf{x})} \left[\log \frac{p_{model}(\mathbf{x})}{p_{data}(\mathbf{x}) + p_{model}(\mathbf{x})} \right] \right)$$

$$= \operatorname{argmin}_G \left(KL \left(p_{data}(\mathbf{x}) \parallel \left(\frac{p_{data}(\mathbf{x}) + p_{model}(\mathbf{x})}{2} \right) \right) + KL \left(p_{model}(\mathbf{x}) \parallel \left(\frac{p_{data}(\mathbf{x}) + p_{model}(\mathbf{x})}{2} \right) \right) + 2 \log 2 \right)$$

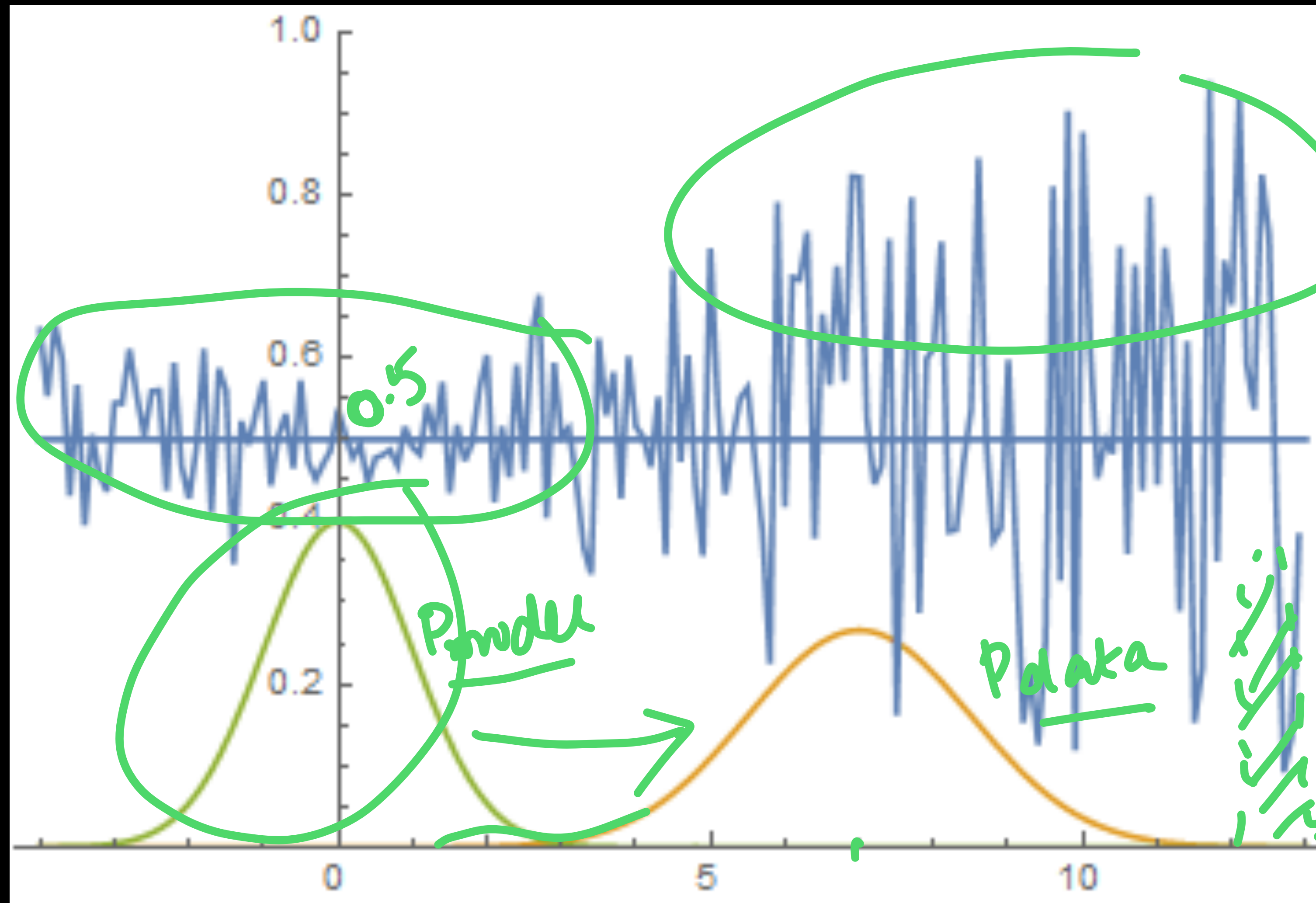
$$= \operatorname{argmin}_{p_{model}(\mathbf{x})} \mathbb{D}_{JS}(p_{data}(\mathbf{x}) \parallel p_{model}(\mathbf{x}))$$

$p_{data} = p_{model}$

Jensen-Shannon divergence.



GAN - Simple example



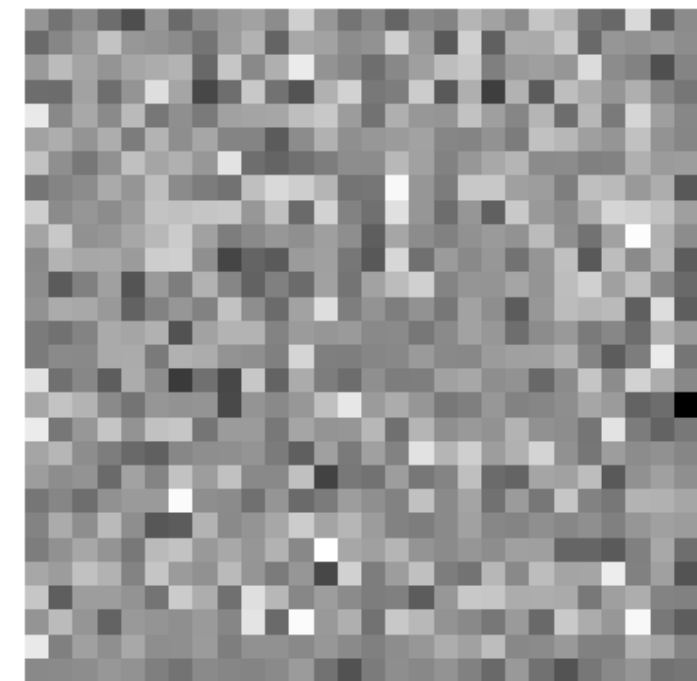
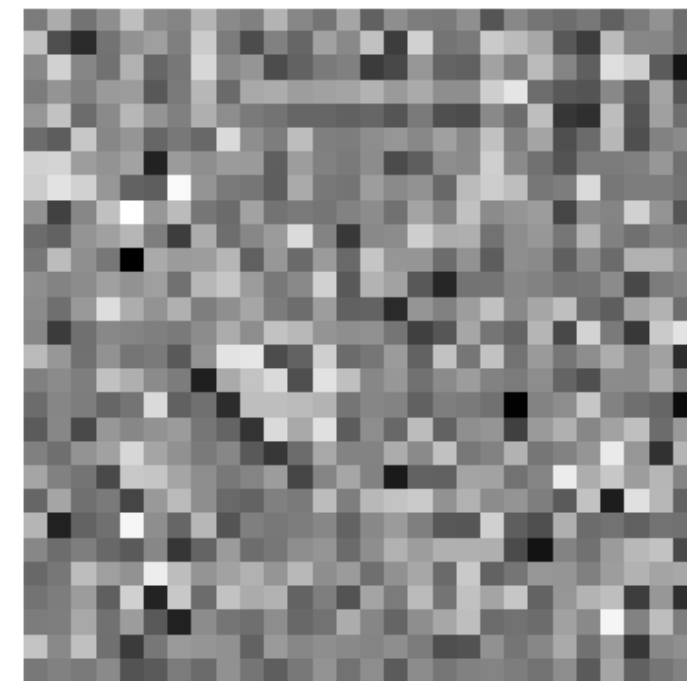
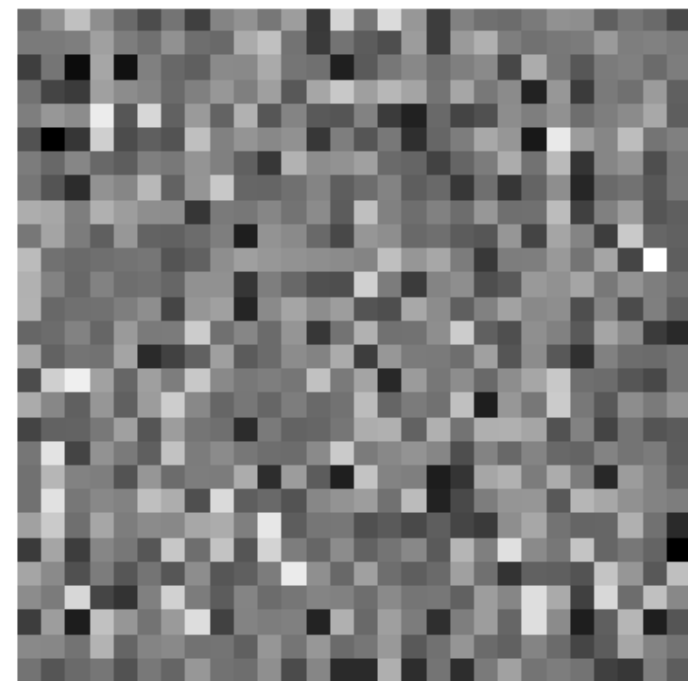
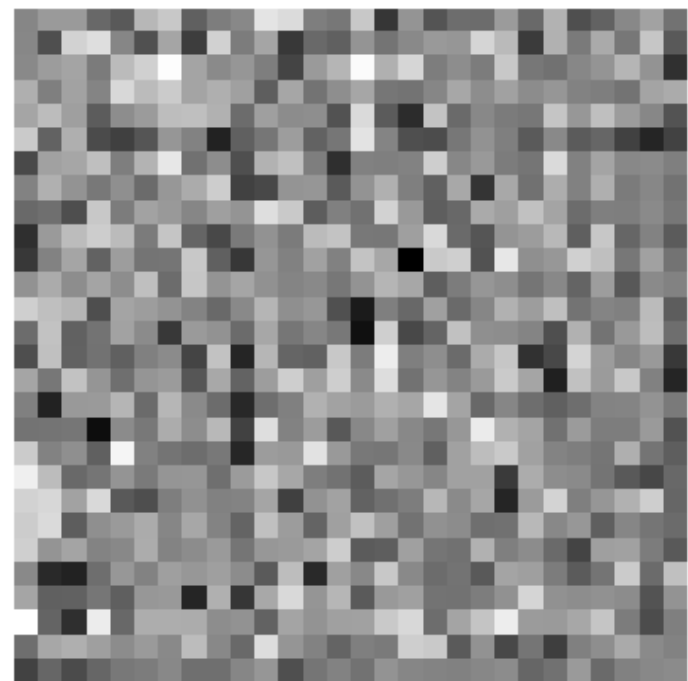
— discriminator
o/r

- Density fn of generated



Generative adversarial networks - examples

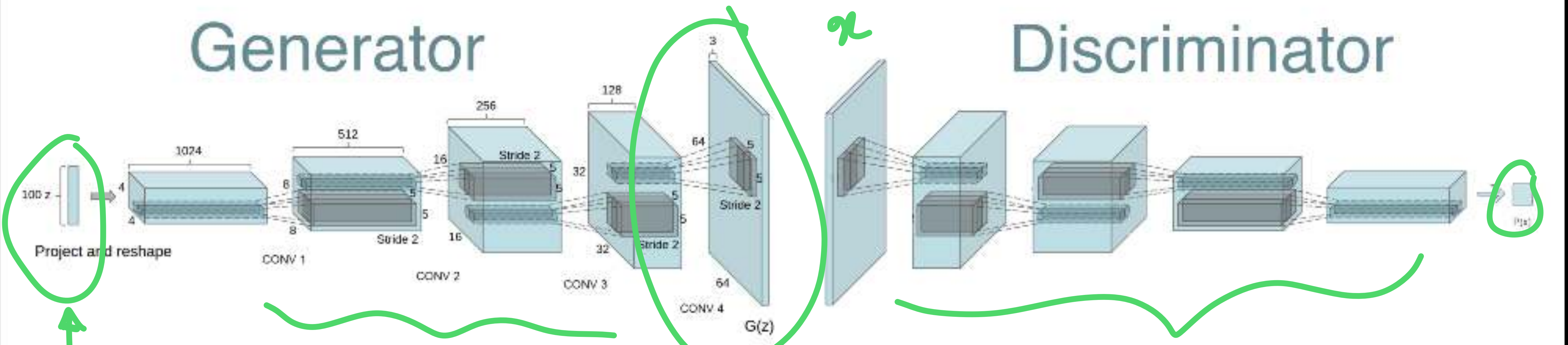
✳ MNIST digits



$$\boxed{D(x) \approx \frac{1}{2}}$$

Deep convolutional GANs (DCGANs)

Generator



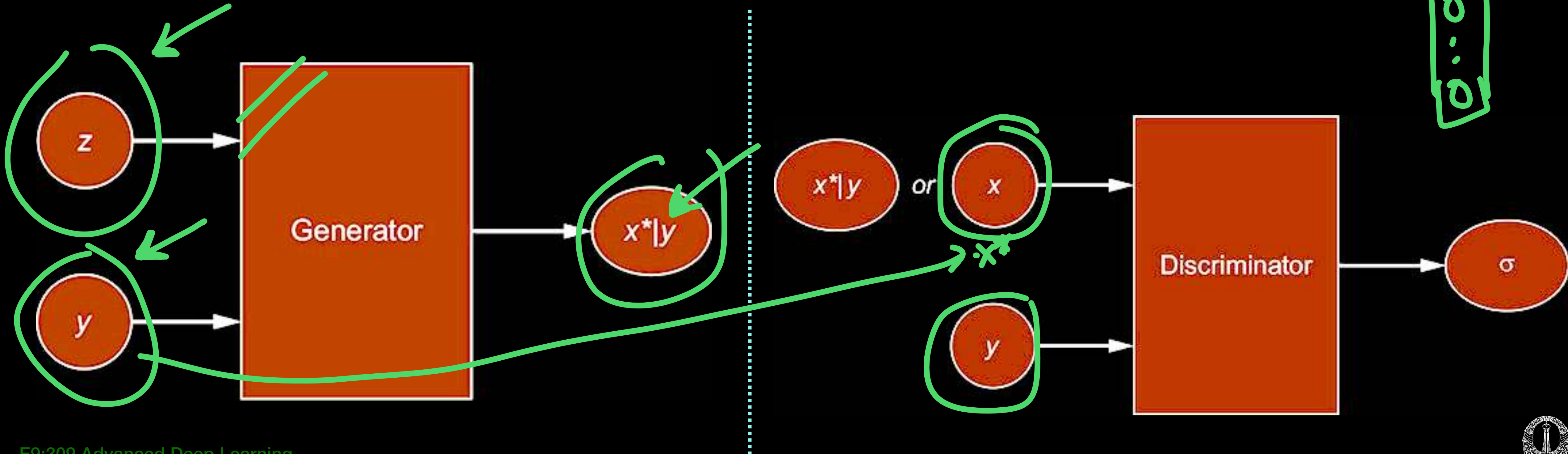
deconvolution

x
 x'

convolution

Conditional GANs

- ✳ Give the generator one hot encodings of classes (features)
 - ➡ This can also contain some attributes of the image.
 - ➡ Model learns the association between the images and the labels

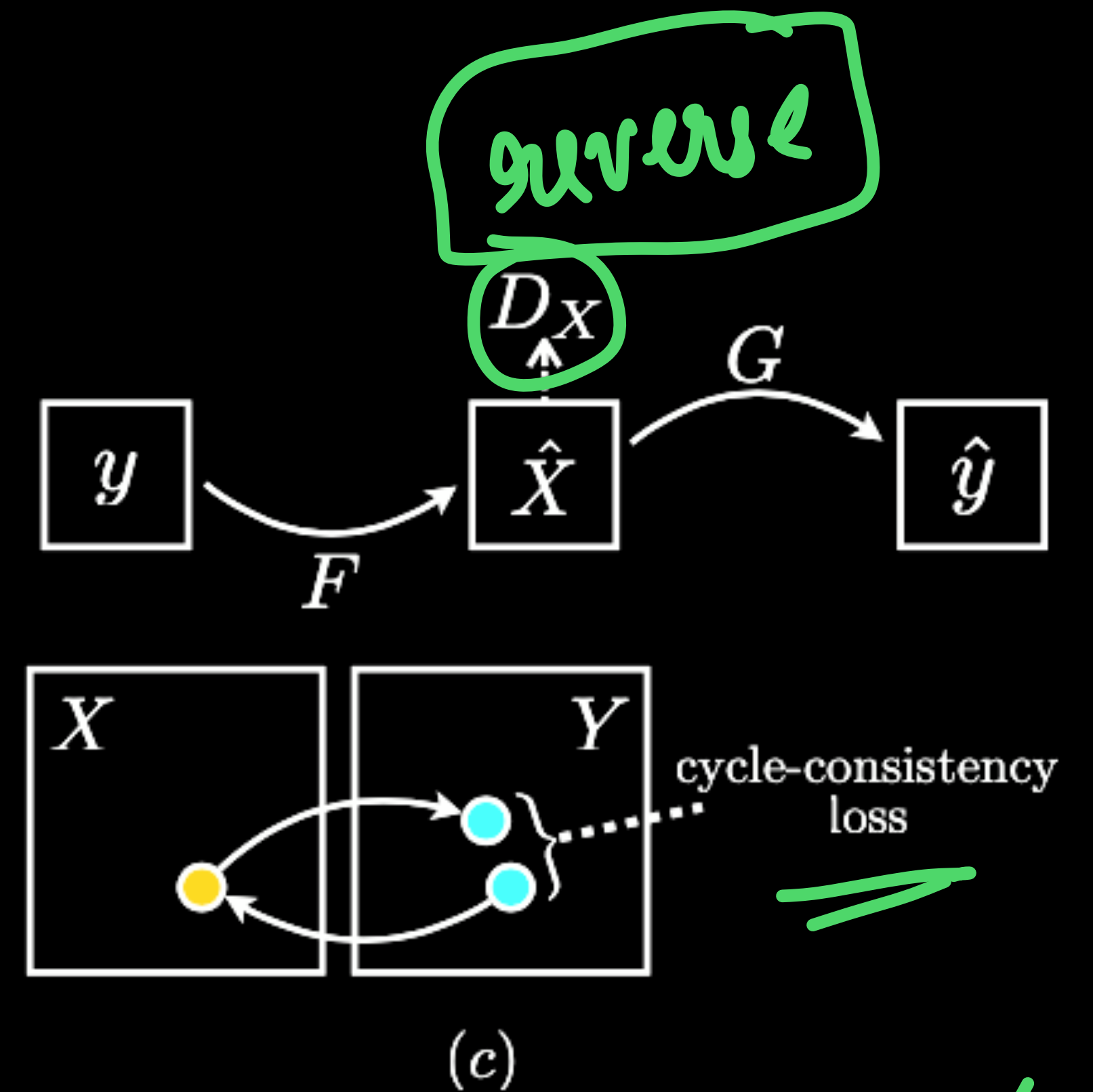
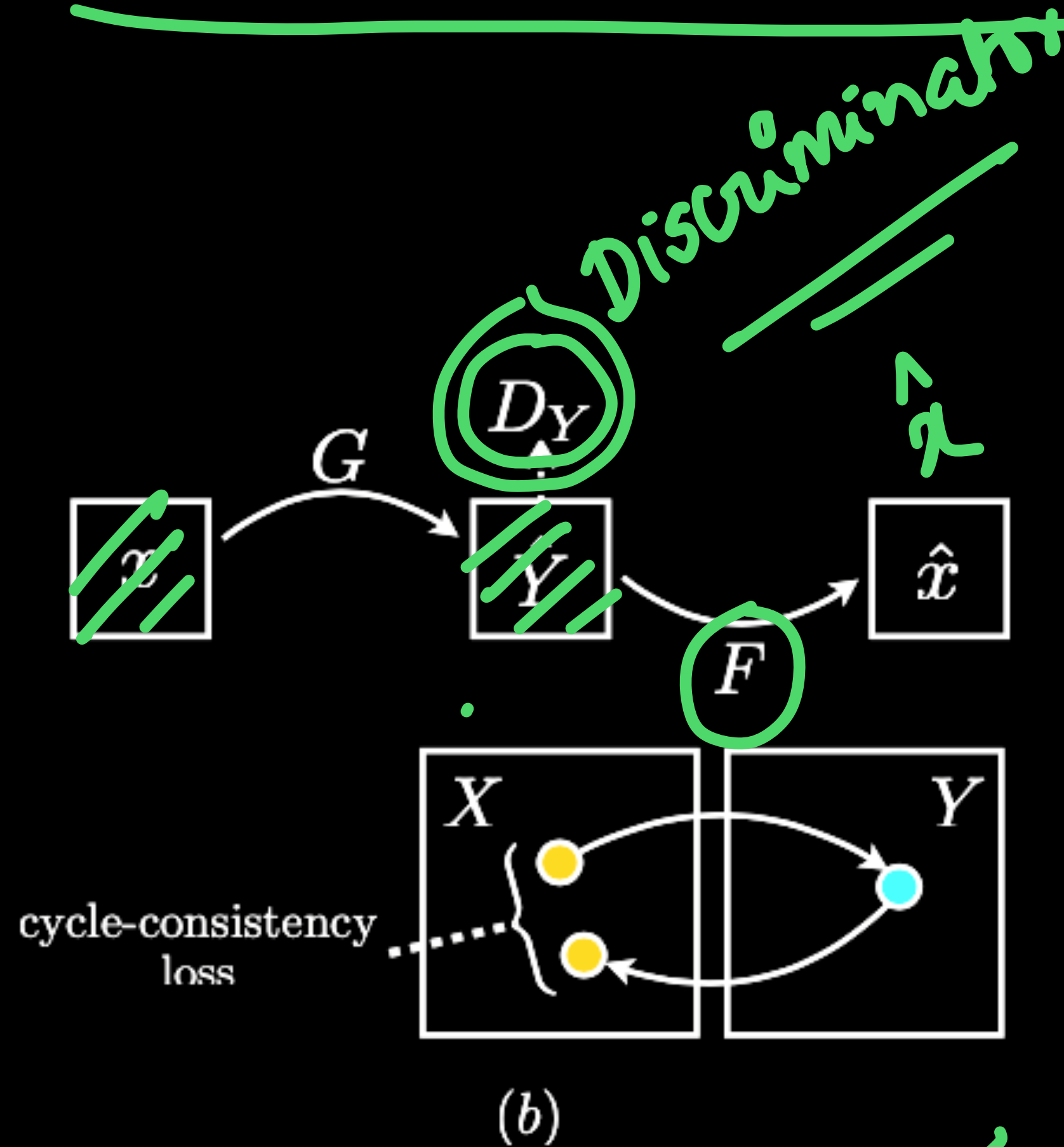
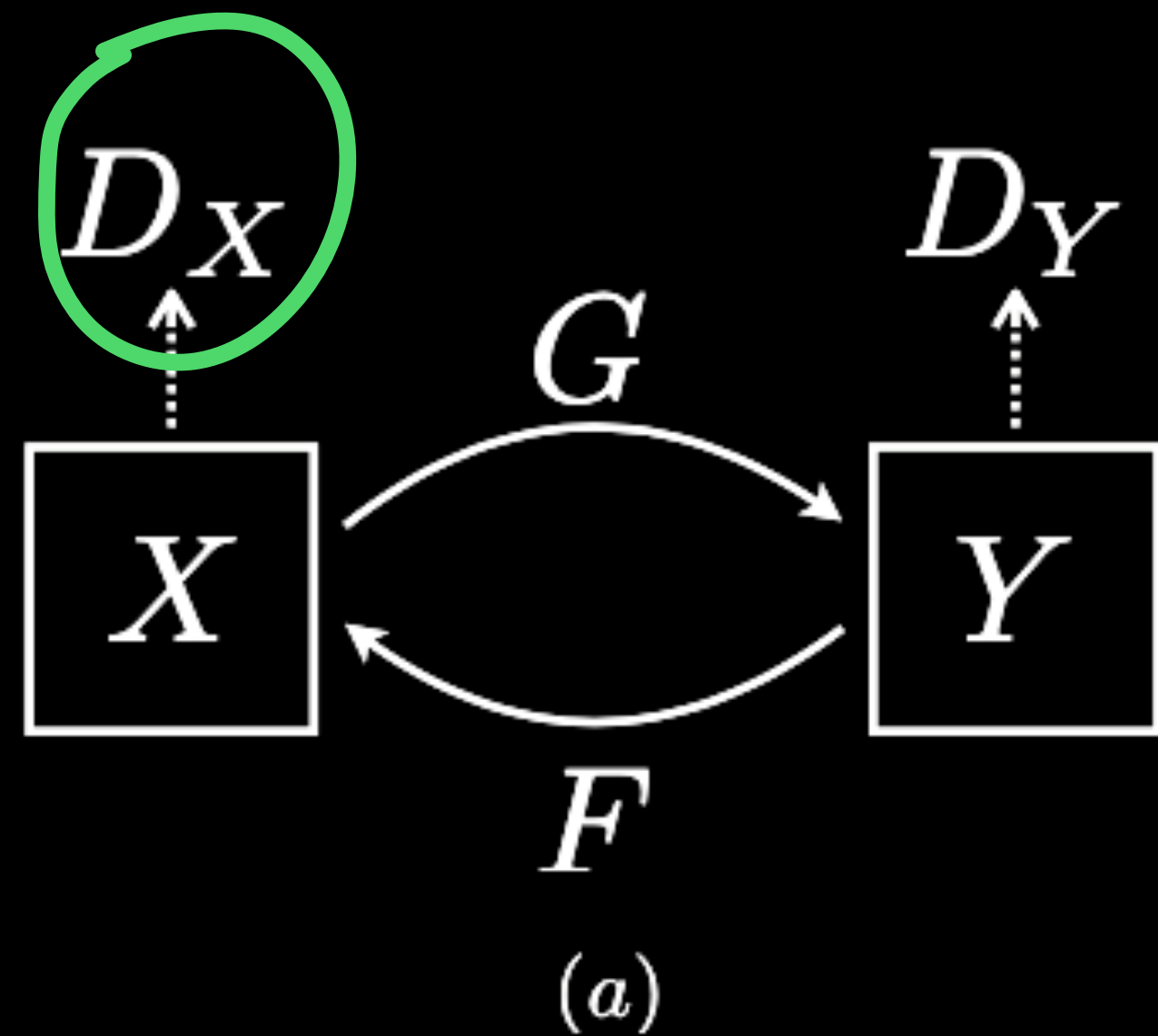


Conditional GANs



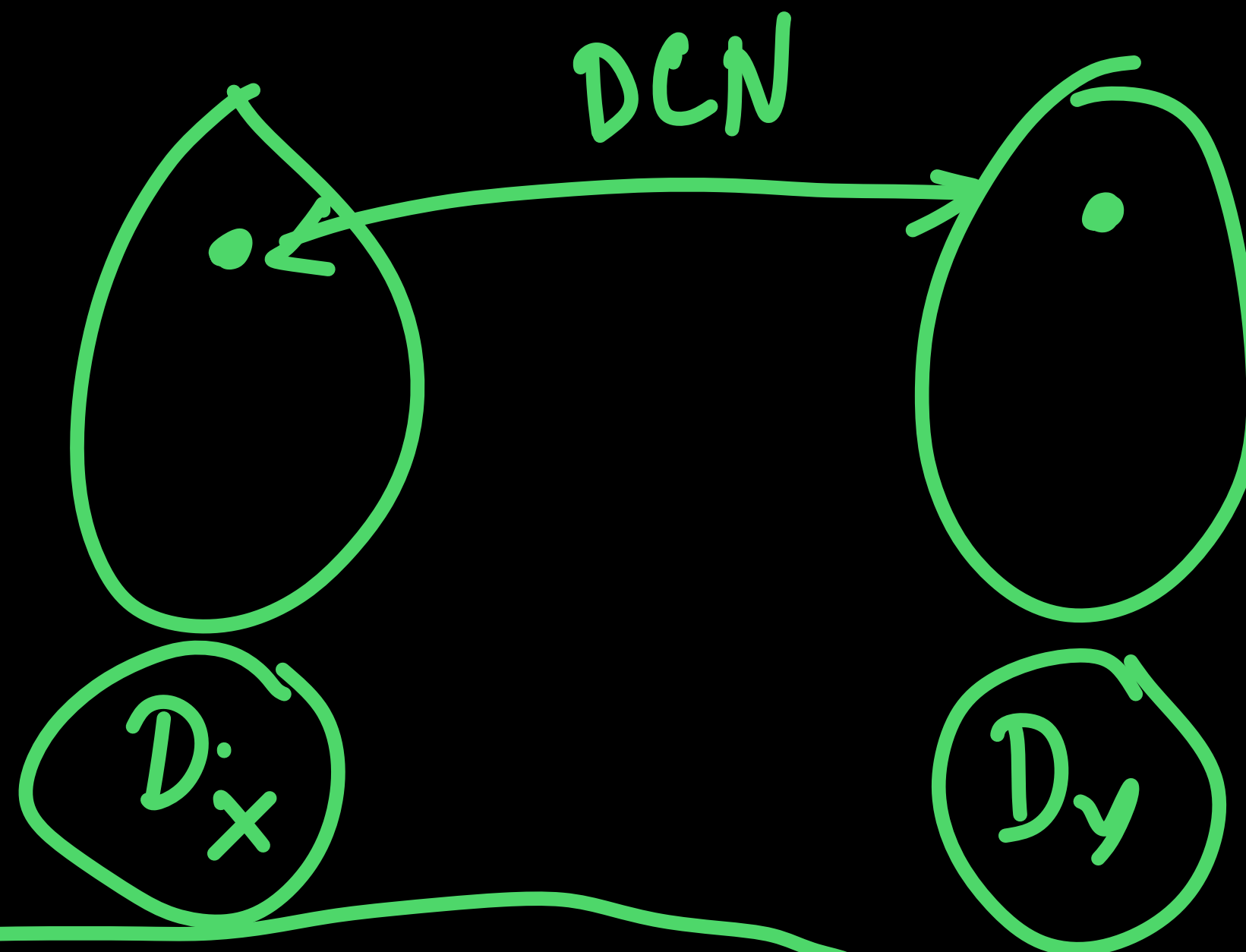
2
0

Cycle GANs - Image to Image translation

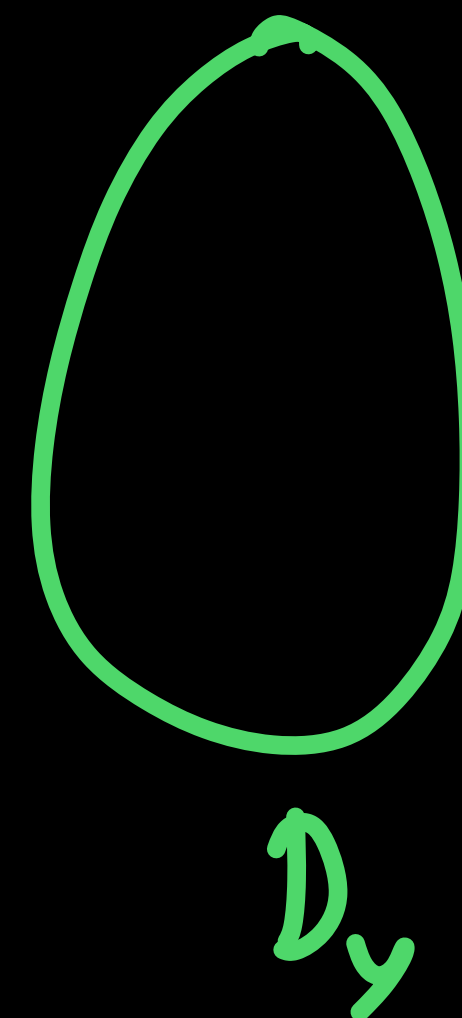
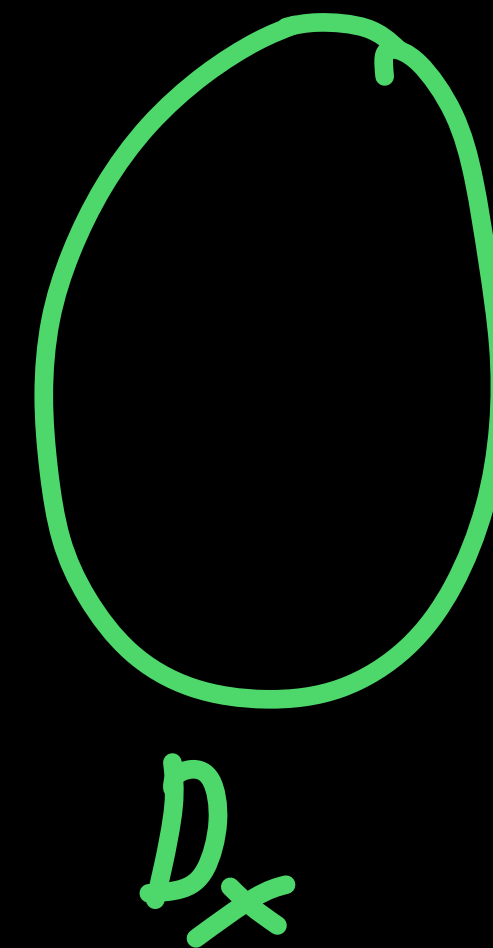


D_X , D_Y

Paired Translation



Unpaired translation

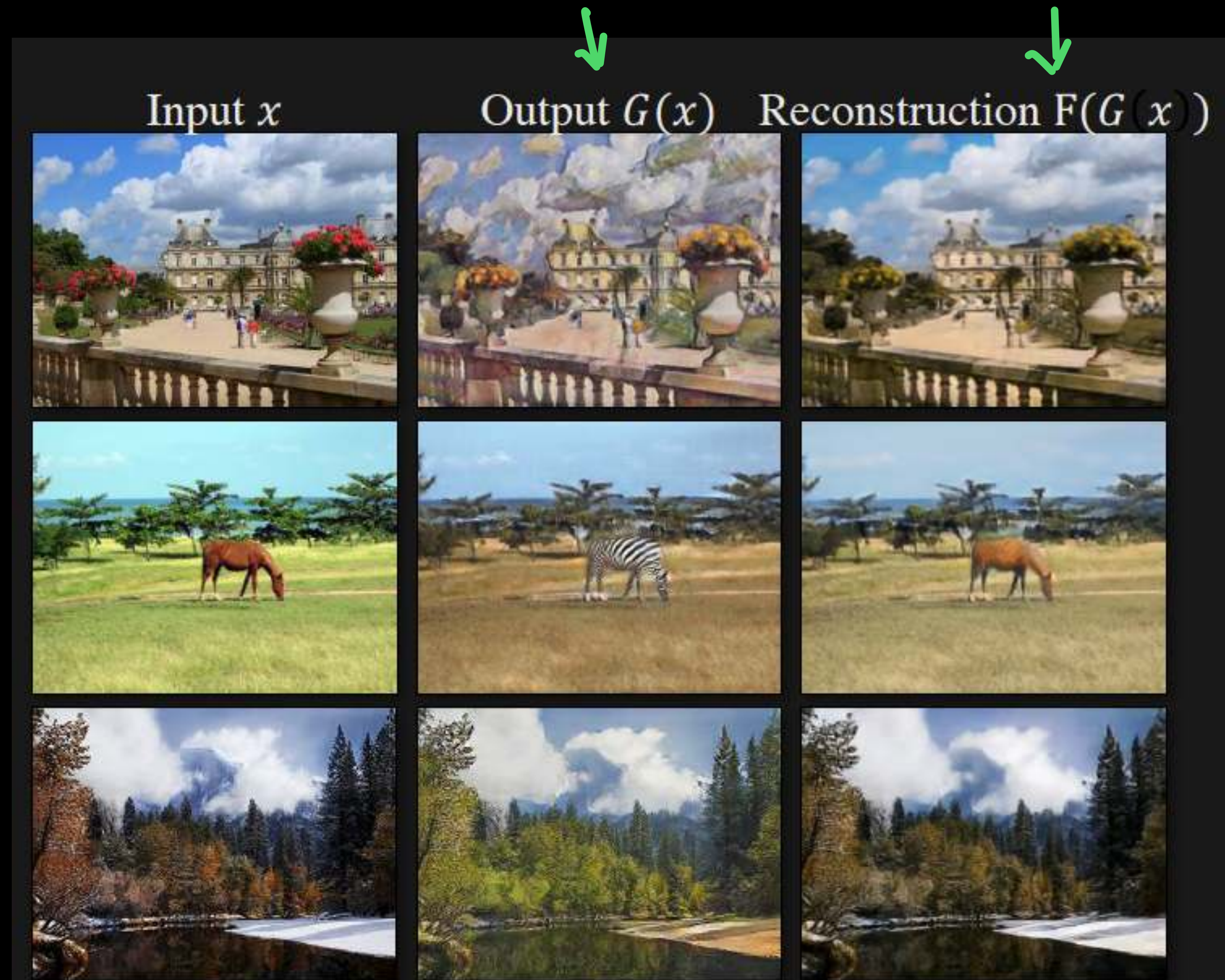


Cycle GANs

painting style →

horses →
to zebras

season change →



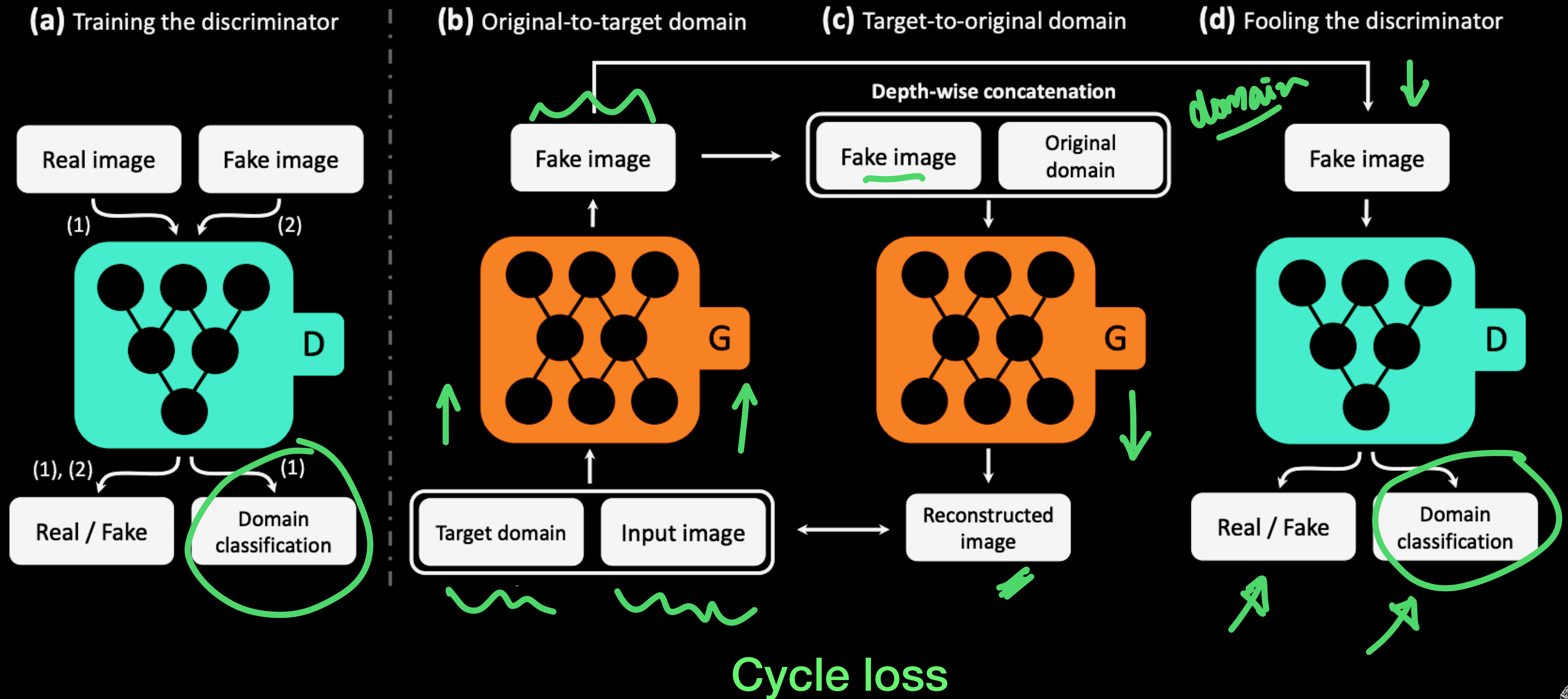
Cycle GANs - test data



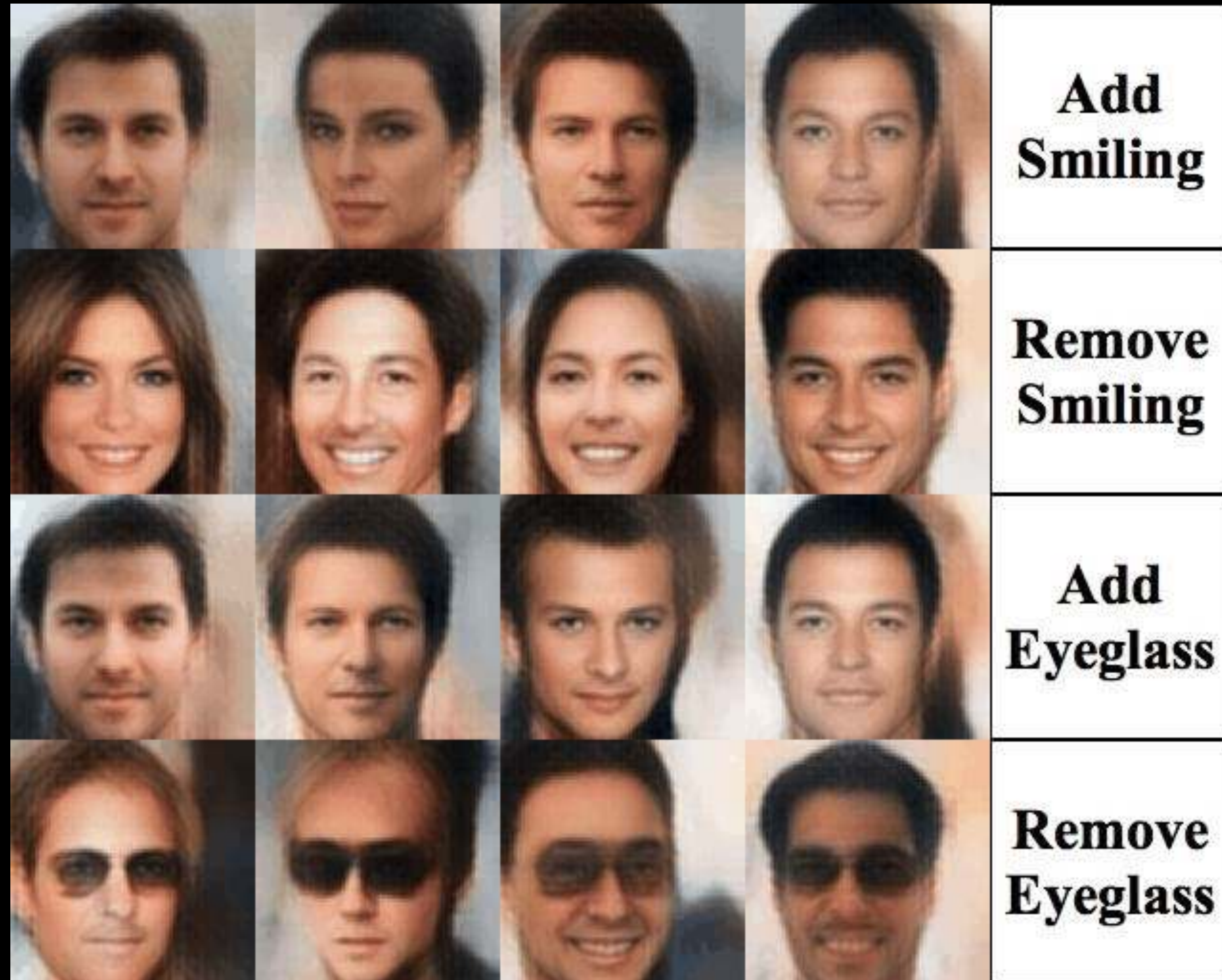
Cycle GANs - test data



StarGANs



StarGANs



Comparing unsupervised learning methods

	RBM	VAE	GAN
DISTRIBUTION	EXPLICIT	EXPLICIT CONDITIONAL DISTRIBUTION	IMPLICIT
LEARNING	<u>APPROXIMATE ML</u>	<u>APPROXIMATE ML</u>	DIRECT OPTIMIZATION
LATENT VARIABLE	BINARY	<u>REAL</u>	<u>REAL</u>
QUALITY	OKAY	MODERATE	GOOD



Topics thus far ...

Visual and Time Series Modeling: Semantic Models, Recurrent neural models and LSTM models, Encoder-decoder models, Attention models. ✓

Representation Learning, Causality And Explainability: t-SNE visualization, Hierarchical Representation, semantic embeddings, gradient and perturbation analysis, Topics in Explainable learning, Structural causal models.

Unsupervised Learning: Restricted Boltzmann Machines, Variational Autoencoders, Generative Adversarial Networks. ✓

New Architectures: Capsule networks, End-to-end models, Transformer Networks.

Applications: Applications in in NLP, Speech, Image/Video domains in all modules.

